

LRT-4051 IND-Enabling Development Plan

Loblolly Ridge Therapeutics | LRT-4051 Program Team | 2026-09-04 | v1.3

LRT-4051 is an investigational oral small molecule for diabetic kidney disease, nominated as a development candidate at gate G2 in September 2025. This plan sets the target date for its IND, the IND-enabling workstreams in dependency order, the partners doing the work, the regulatory interaction plan, the risks to the timeline and what the G3 decision will need. Version 1.3 updates the forecast after the August schedule review.

Target: IND submission June 4, 2027

The FY2026 plan baseline was April 30, 2027. The forecast is now June 4, 2027, because GLP toxicology final reports are forecast for December 4, 2026 (baseline October 30) and GMP API release for December 11, 2026 (baseline October 16). Both sit on the critical path. The IND-enabling workplan has 44 tasks: 13 complete, 10 in progress and 21 not started, with 18 on the critical path.

Workstreams in dependency order

Nonclinical safety. GLP repeat-dose toxicology in two species drives the plan. The in-life phases finished in late July, about two weeks after plan; draft reports follow, then an internal QA audit of the drafts, then final reports. ICH M3(R2) links the duration of repeated-dose toxicity studies to the duration and scope of the planned clinical trials, so the study durations were set for the planned first-in-human study. The genetic toxicology battery is complete and the safety pharmacology core battery is due in October.

DMPK. Toxicokinetic sample analysis feeds the human dose projection memo, which the first-in-human synopsis and the nonclinical summaries both need. Metabolite profiling in the toxicology species runs in parallel.

CMC. The GMP API campaign is followed by release testing and a certificate of analysis, then clinical drug product manufacture and release. The API stability program starts after release.

Regulatory writing. Nonclinical written and tabulated summaries start when the final toxicology reports arrive; the quality module and investigator's brochure run alongside; IND assembly and publishing follow.

Who does the work

Quarry Hill Preclinical runs GLP toxicology and safety pharmacology, Ashcombe Genetic Toxicology the genetic toxicology battery, Alder Bioanalytical GLP bioanalysis and Kestrel DMPK Services non-GLP pharmacokinetics. Tessaro Fine Chemicals makes GMP API and Northgate Pharma Services drug product; Brookmoor Analytical does analytical and stability work. Inkwell Medical Writing and Parkway Regulatory Publishing support the IND documents. Each GLP testing facility has a quality assurance unit independent of the study staff, which our qualification confirms, and our QA audits the draft toxicology reports.

Regulatory interaction plan

We will request a pre-IND meeting with FDA, now forecast for November 20, 2026, with the meeting expected in February 2027. The briefing package will cover the nonclinical program, the CMC approach and the first-in-human design. Advice from the meeting will be reflected in the IND and in the final first-in-human protocol.

Risks to the timeline

Toxicology reporting is the main risk. Any further delay in draft reports moves the QA audit, the final reports and the nonclinical summaries day for day.

API release depends on the campaign at Tessaro and release testing at Brookmoor; the genotoxic impurity method must be ready before release testing.

First-in-human sourcing has not been decided, and the request for proposal planned for November and December could run into January. This affects study start, not the IND date.

The regulatory team also supports the LRT-1107 program in 2027, so writing capacity will be tight in the first half of the year. Study titles and report numbers for all IND-enabling studies are listed in Appendix A.

The G3 decision

The G3 decision to submit the IND is planned for April 2027 and will be made by the CEO on the Portfolio Review Committee's recommendation. Gate G3 asks whether the IND package is complete and the first clinical study is ready to place. The gate package will show the complete nonclinical and CMC evidence, the pre-IND outcome, the first-in-human plan and sourcing decision, and the 2027 budget and FTE request. Under the stage-gate procedure the CEO approves G0 to G3 decisions within budget.